AI-Powered Data Analysis Internship – Final Consolidated Report

Week 4

Churn Prediction, EDA, Recommendation System & Strategic Insights

Team Name:1011 – AI Group 10

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Executive Summary

This project represents the culmination of four weeks of structured analytical work during the AI-Powered Data Analysis Internship. Using a real-world dataset of student engagement, applications, and program participation, our team developed a full end-to-end analytics pipeline covering:

- **Data cleaning & feature engineering**
- **Exploratory Data Analysis (EDA)**
- **Churn prediction modelling**
- **Insight extraction & strategic recommendations**
- **A basic recommendation system for engagement improvement**
- **Final presentation & reflection preparation**

Across all phases, the primary goal was to understand the factors contributing to student churn (drop-out/withdrawal) and build actionable, data-driven solutions. The final Gradient Boosting model achieved a strong **ROC−AUC ≈ 0.93**, demonstrating high predictive ability. Analysis also revealed clear behavioral and program-related drivers of churn, which informed a practical recommendation system and targeted retention strategies.

This report consolidates the full analytical journey, major findings, visualizations, and actionable next steps.

1. Introduction

Student churn poses significant operational, financial, and strategic challenges for academic programs. Identifying high-risk individuals early enables proactive intervention, improved retention, and optimized program outcomes.

The project aimed to:

1. Clean and validate the dataset to ensure reliability.
2. Understand student behavior and engagement patterns.
3. Build predictive models that flag students at risk of churn.
4. Extract insights to guide program improvement.
5. Create a simple recommendation system to support engagement.
6. Present all findings in a clear, stakeholder-ready presentation.

The dataset consisted of ~8.5k records with demographic, engagement, and program-related variables. The target variable (**churn**) was defined using status fields, enabling binary prediction.

2. Data Preparation & Feature Engineering

2.1 Dataset Overview

- Source: production_ready_dataset_v3_validated.xlsx
- Final row count after validation: **8,558**
- Target variable:
  - **1 = Dropped Out / Withdraw**
  - **0 = All other statuses**
- Class imbalance:
  - Churn = **8.2%**
  - Non-churn = **91.8%**

2.2 Cleaning & Fixes Performed

- Removed leakage fields (status_*, raw dates after feature extraction).
- Fixed **engagement inversion errors** by converting invalid timelines to NaN.
- Standardized date formats and engineering fields such as:
  - signup_year, signup_month
  - engagement_lag_days_fixed
  - opportunity_duration_days
  - days_before_start
  - engagement_lag_bucket
- Converted categorical fields to proper string types.

2.3 Missing Value Handling

- **Numeric:** Median imputation
- **Categorical:** Most frequent imputation

2.4 Encoding & Scaling

All transformations were implemented via a **Pipeline + ColumnTransformer**, ensuring reproducibility and preventing leakage.

3. Exploratory Data Analysis (Week 2)

3.1 Distribution Insights

The dataset revealed:

- Most students fall in the **21–25 age range**.
- Engagement lag shows **heavy right skew**, with some extremely delayed applicants.
- Opportunity duration ranges from **0 to ~700 days**, depending on category.

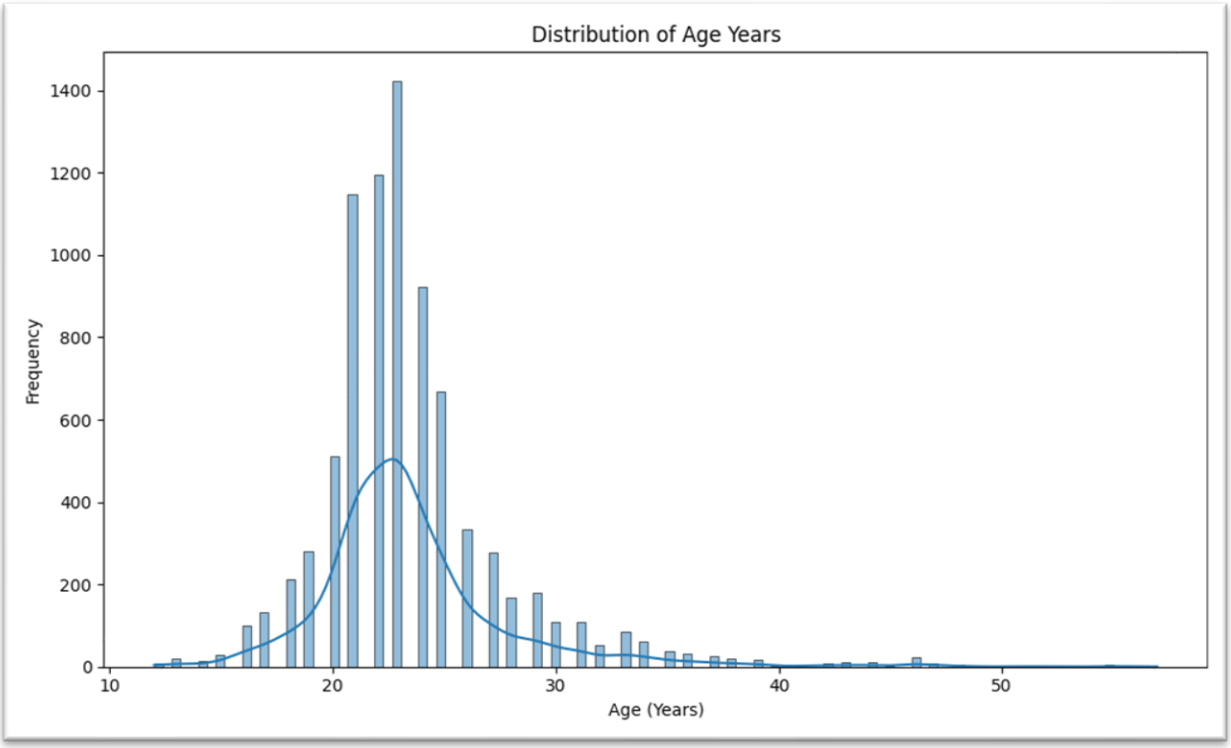

Figure 1: Age Distribution

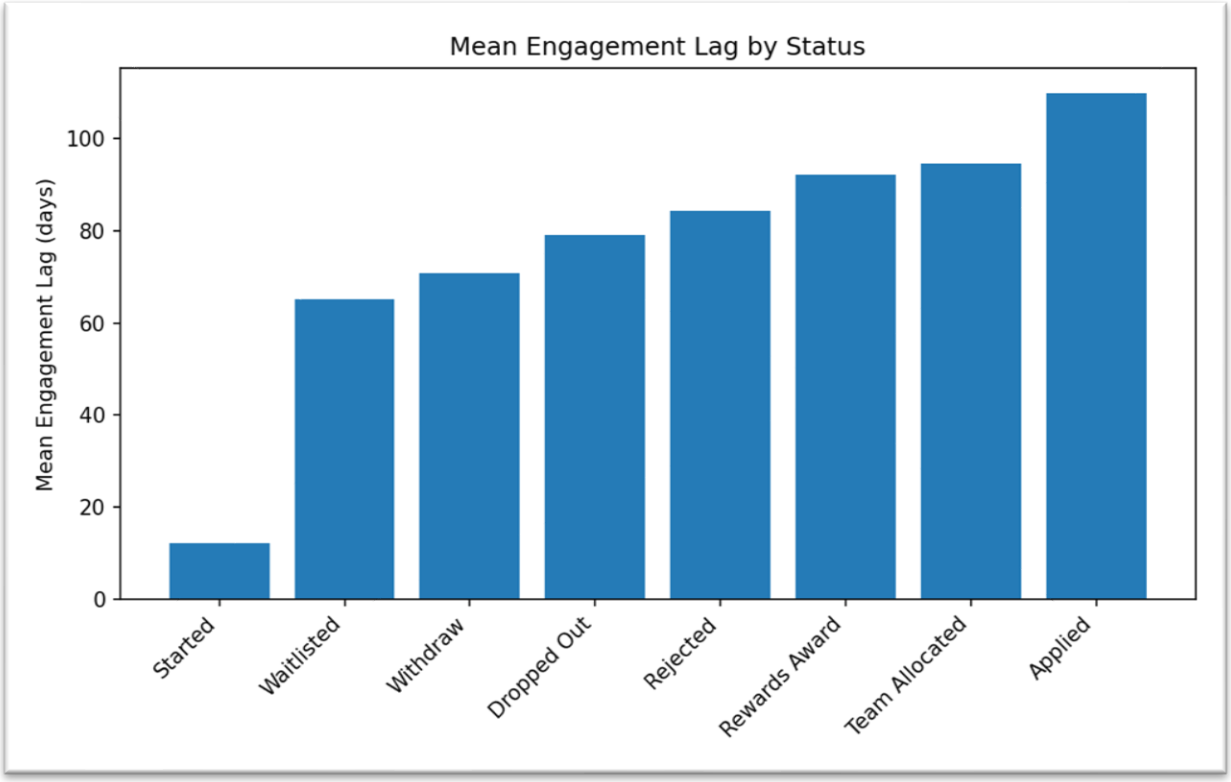

Figure 2: Engagement Lag Boxplot

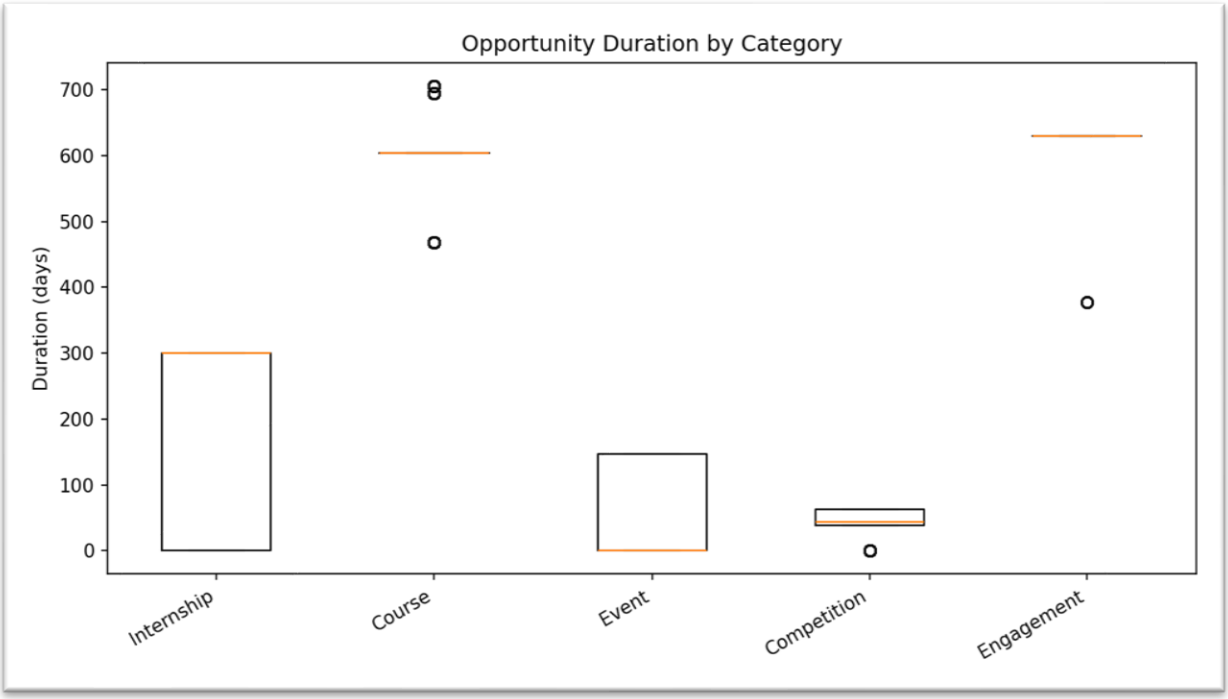

Figure 3: Opportunity Duration Boxplot

3.2 Signup and Status Trends

- Signups peak in **mid-year months**, especially July–September.
- Majority statuses:
  - Rejected (41.7%)
  - Team Allocated (38.3%)
- Churn categories:
  - Dropped Out
  - Withdraw

3.3 Numeric & Behavioral Patterns

Key observations:

- Longer **engagement lag** correlates with higher churn likelihood.
- Students who engaged **after program start** had higher dropout probability.
- Zero-duration opportunities indicate administrative or structural inconsistencies.

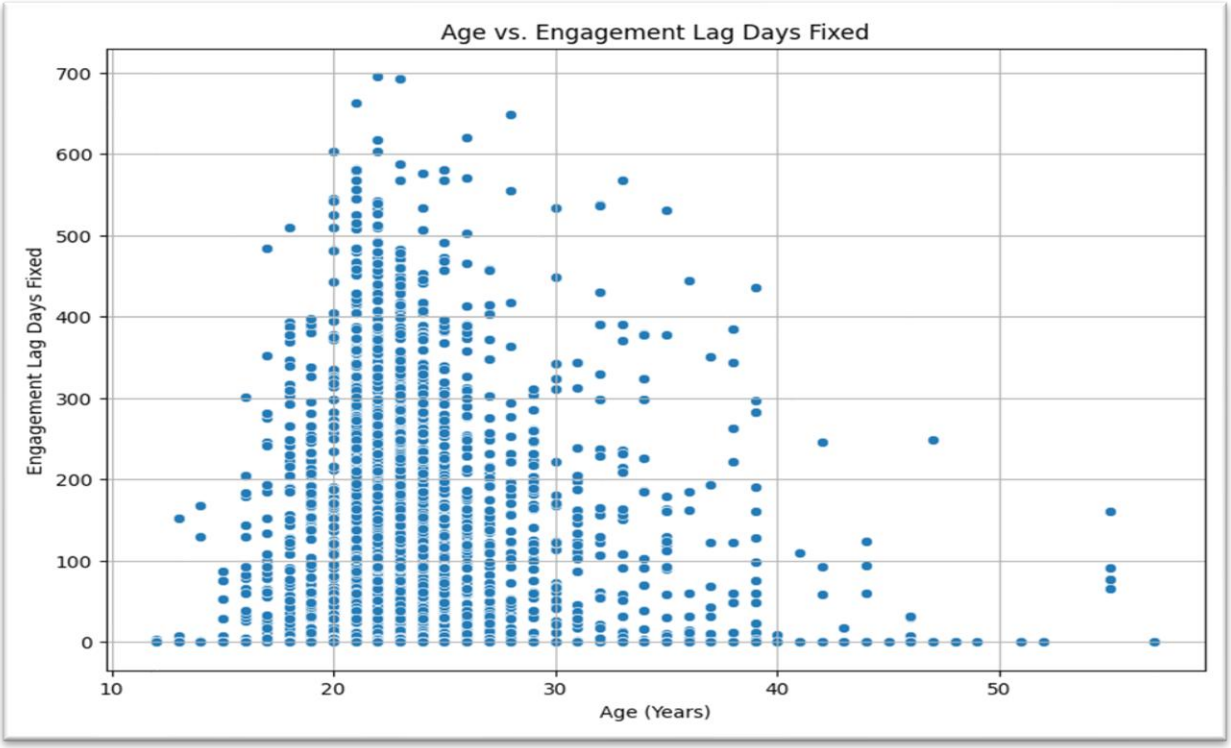

Figure 6: Scatter Plot – Age vs Engagement Lag

4. Churn Prediction Modeling

4.1 Models Trained

- Logistic Regression
- Random Forest
- Gradient Boosting (best performer)

4.2 Evaluation Metrics

Model	ROC-AUC	PR-AUC	Strength
Gradient Boosting	**0.927**	0.566	Best discriminator
Random Forest	0.901	0.498	Strong baseline
Logistic Regression	0.878	0.383	High recall, low precision

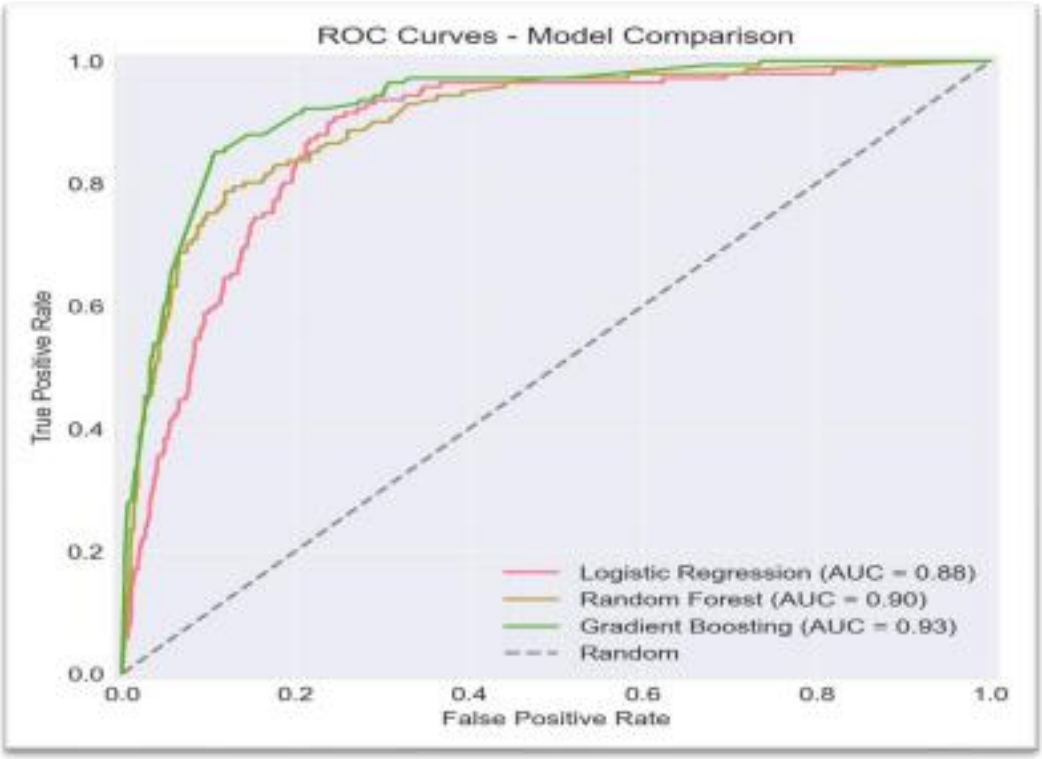

Figure 8: ROC Curves for All Models

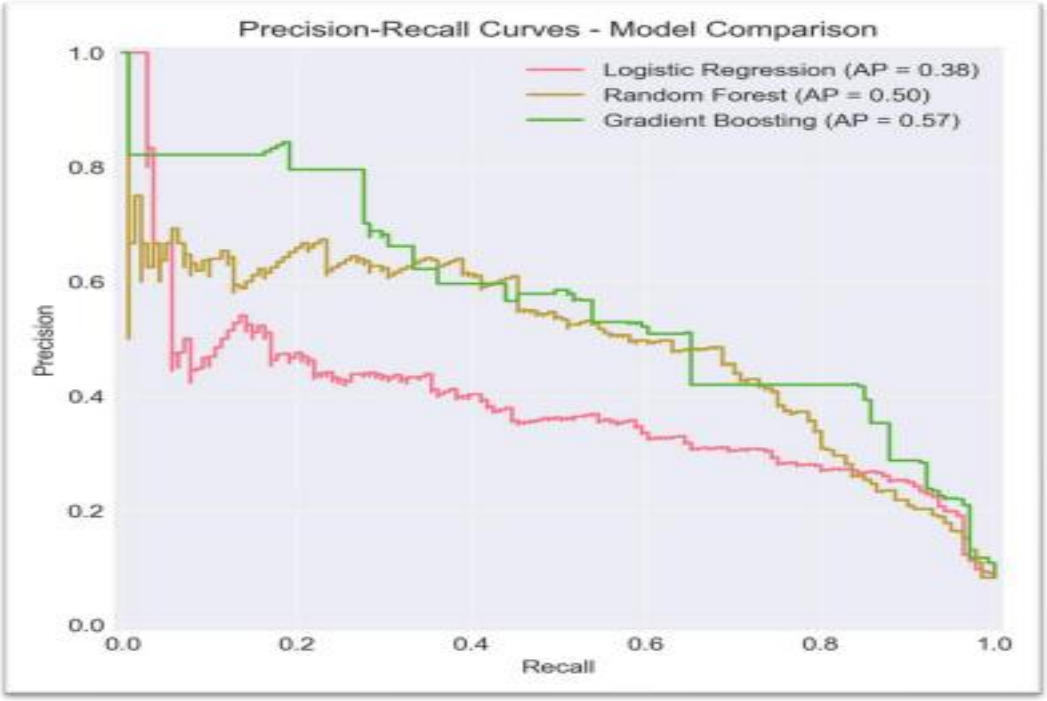

Figure 9: Precision–Recall Curves

4.3 Confusion Matrix (Best Model)

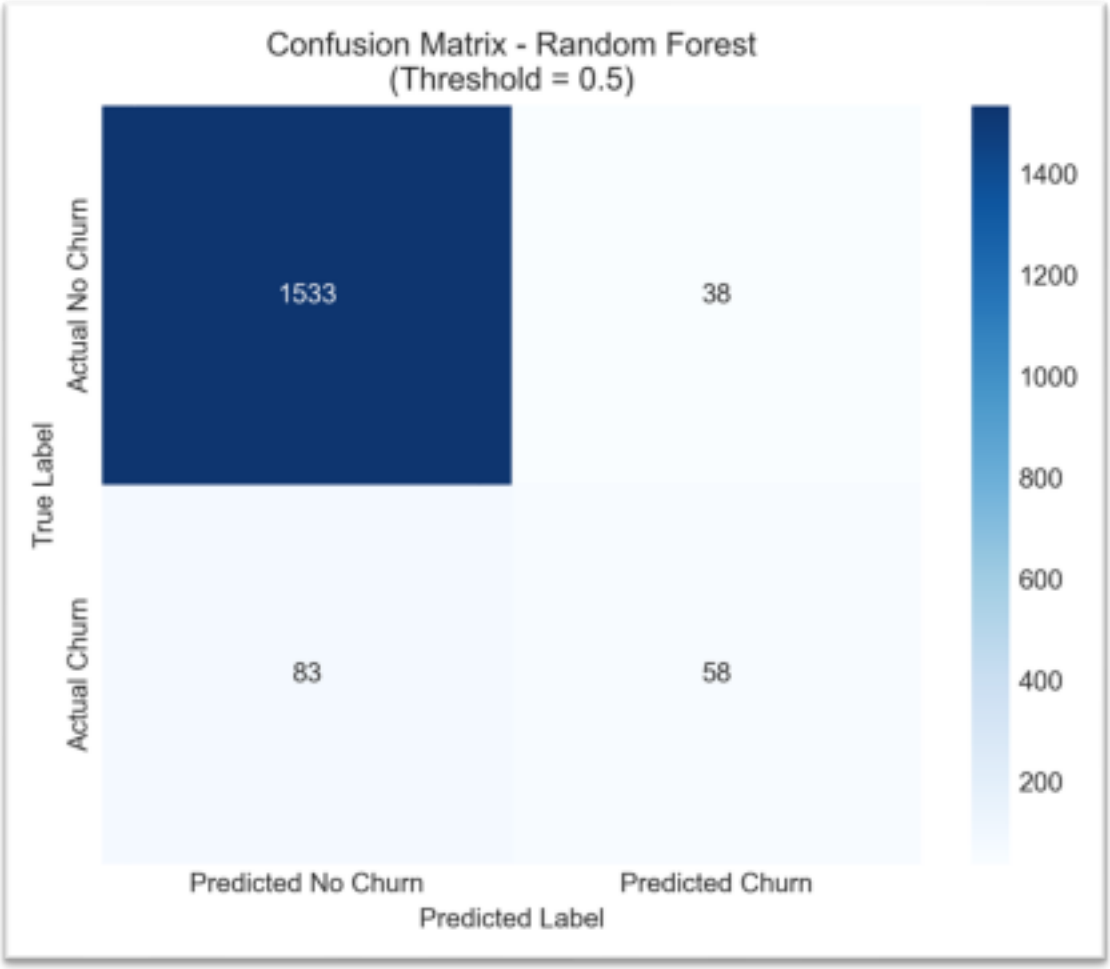

Figure 10: Confusion Matrix – Gradient Boosting

4.4 Feature Importance (Drivers of Churn)

Top influential factors:

- **Engagement Lag (fixed & bucket)**
- **Opportunity Category**
- **Days Before Start**
- **Opportunity Duration**
- **Country / Institution**

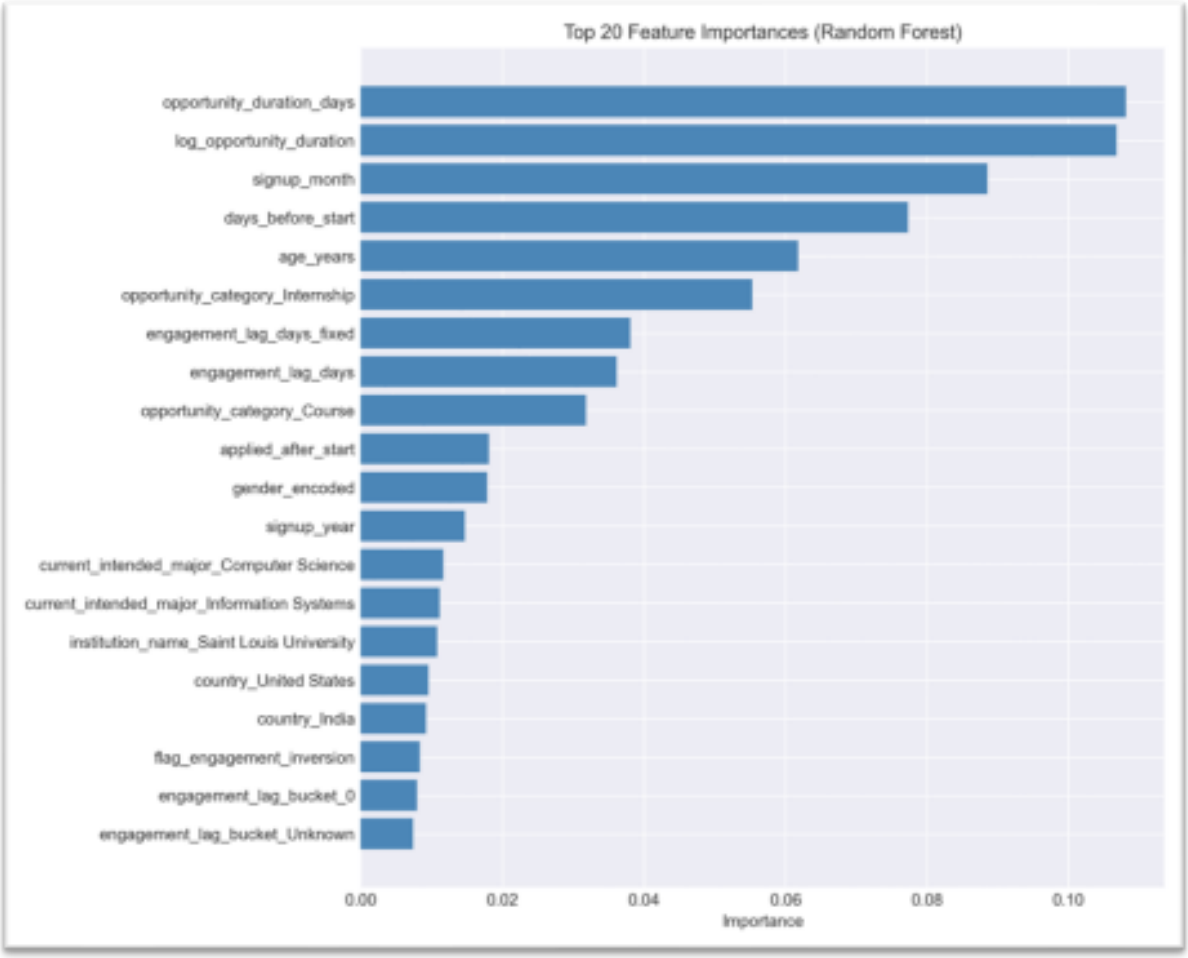

Figure 11: Feature Importance – Top 20 Features

Insights clearly show that timing, engagement, and program characteristics drive churn far more than demographics alone.

5. Recommendations (Data-Driven Strategies)

Based on EDA + predictive modeling:

5.1 Improve Early Engagement

- Automatically flag students with **lag > 7 days**.
- Trigger reminders within **24–48 hours of application**.

5.2 Strengthen High-Churn Opportunity Categories

- Review curriculum intensity.
- Provide structured onboarding.
- Offer early mentorship within the first week.

5.3 Address High-Risk Countries/Institutions

- Localized time-zone friendly support.
- Accessible learning resources for low-bandwidth regions.
- Coordinated institution-specific collaboration.

5.4 Program Duration-Based Interventions

- Long-duration programs need **milestone tracking**.
- Short-duration programs need **rapid engagement nudges**.

6. Recommendation System

A simple but effective **Content-Based + Rule-Based Hybrid** system was built.

6.1 System Logic

Uses:

- Engagement lag
- Opportunity category
- Student interest (major)
- Previous churn risk signals
- Program duration

Outputs include:

- **Personalized program recommendations**
- **Engagement improvement suggestions**
- **At-risk alerts with reason codes**

7. Conclusion

Across the internship, we built a fully functional analytical pipeline that:

- Cleaned and validated a complex dataset
- Performed deep EDA and pattern discovery
- Built high-performing churn prediction models
- Identified actionable churn drivers
- Developed a recommendation system

- Integrated insights into a cohesive presentation

The final model, visualizations, and recommendations provide a practical foundation for improving student retention in real-world scenarios. This project demonstrates strong analytical, modeling, and communication skills aligned with industry expectations for AI Data Analysts.